

ISC Telemetry Operational Guide

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Hardware Setup, ISCmetrics Suite & Marple Cloud Data Platform

1. Physical Hardware Setup

1.1 Car Side (Transmitter / ECU)

- **Antenna Positioning:** Mount vertically on the roll hoop / shark fin with an insulating spacer. Maintain at least 3.5 cm ($\lambda/4$) clearance from carbon fiber panels to prevent RF shadowing.
- **RF Cabling:** Use low-loss RG-316 or RG-142 coaxial cable with SMA connectors (< 50 cm). Route away from high-voltage (HV) cables and inverter phase lines.
- **Power Decoupling:** High-power PA modules draw up to 150 mA in peak bursts. Solder a 10–100 μ F electrolytic in parallel with a 100 nF ceramic capacitor directly across the module's VCC (3.3V) and GND pins.

1.2 Pitwall Side (USB Receiver Ground Station)

- **Hardware:** Arduino Nano / USB dongle connected to nRF24L01+ + PA/LNA module over SPI, communicating via USB-Serial at 115200 baud.
- **Mast Elevation:** Mount the high-gain antenna (+9 to +14 dBi) on a tripod mast 2.5 m to 3.5 m above ground to ensure line-of-sight and clear the Fresnel zone.

2. Installing & Running ISCmetrics

ISCmetrics is the official Formula Student telemetry desktop application developed for real-time monitoring, state machine diagnostics, and CSV logging.

- **GitHub Repository:** <https://github.com/MrAndy5/ISCmetrics>
- **Releases / Pre-built Binary:** <https://github.com/MrAndy5/ISCmetrics/releases/latest>

2.1 Installation from Source

```
git clone https://github.com/MrAndy5/ISCmetrics.git
cd ISCmetrics
python -m venv venv
.\venv\Scripts\Activate.ps1
pip install PyQt5 matplotlib pyserial numpy pandas marple requests python-can
python ui.py
```

2.2 Cloud Upload Authentication

★ MARPLE CLOUD UPLOAD PASSWORD:

To enable cloud uploads in the Settings dialog, enter the password:

Password: ISC_telemetry_2026

Workflow: Click Settings (Gear Icon) -> Check 'Upload to Marple Data (cloud)' -> Enter Password -> Save.

2.3 Live Session Workflow

1. Select the Serial COM Port for the receiver dongle in the top bar dropdown.
2. Click **CONNECT** to begin live streaming (Inverter FSM, DC Bus, AMS voltages & temps, APPS pedals, Brake pressure, Link health).
3. Click **DISCONNECT** when the run completes. The session CSV is saved under logs/ and automatically pushed to Marple in the background.

3. Marple Data: Cloud Uploads & Analysis

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- **Default API Token:** mdb_Le69BDaNdgn1SJ4DqWX6N6btH-a8Dx8Ou96aBbLA4v8
- **Target Stream:** ISC_Telemetry
- **Local Override:** Add MARPLE_API_TOKEN=your_token to a .env file in the directory.

3.1 Post-Race Data Fusion (SD Card Merge)

1. Open **Tools** -> **Post-Race Analysis** in ISCmetrics.
2. Select the recorded **Session CSV**.
3. Select the car's SD-card **GPS Log** (.nmea) and set the UTC timezone offset.
4. Select the **AMS Temperature Log** (95 cells per-module data).
5. Click **Merge & Push to Marple**. The merged dataset is uploaded instantly.

3.2 Recommended Marple Derived Channels

Channel Name	Formula / Expression	Purpose / Alert
ams_max_cell_mv	MAX(vmax_mod0..4)	Overvoltage alert (> 4200 mV)
v_cell_min_mV	v_cell_min_mV (Raw)	Undervoltage alert (< 3000 mV)
ams_max_temp_c	MAX(tmax_mod0..4)	Thermal ceiling alert (> 55°C)
current_A	corriente_accu * 0.1	Tractive pack current in Amps
seq_jump	seq - PREV(seq)	Radio packet loss (seq_jump > 1)

4. RF Range Optimization & Trackside Cheat Sheet

- **PA + LNA:** +20 dBm (100 mW) TX power + 10 dB LNA gain gives +35 dB to +42 dB total link margin improvement.
- **RF Channel:** Set to Channel 76 or higher (2476 MHz+) to avoid trackside 2.4 GHz WiFi congestion.
- **250 kbps Mode:** Extends receiver sensitivity to -94 dBm for maximum range across large tracks.

1. **PLUG IN:** Connect Arduino USB receiver dongle to laptop.
2. **ELEVATE:** Mount pit antenna on tripod mast (2.5m+ height).
3. **CLONE/RUN:** Clone <https://github.com/MrAndy5/ISCmetrics.git> & run 'python ui.py'.
4. **UNLOCK CLOUD:** Settings -> [X] Upload to Marple -> Enter: **ISC_telemetry_2026**
5. **CONNECT:** Select COM Port -> Click **CONNECT**.
6. **POST-RACE:** Tools -> Post-Race Analysis -> Merge SD GPS/AMS -> Upload to Marple.